Projects Guide

Natural Language Processing in Python — HSE University

Assoc. Prof. Soroosh Shalileh

Spring Semester (Modules 3–4)

Overview

This guide outlines expectations and evaluation criteria for the two major course projects in the *Natural Language Processing in Python* course. Both projects are designed to foster research thinking, technical depth, and clear scientific communication.

- **Mini-Project (Module 3)**: Student-designed midterm project focused on foundational NLP methods.
- **Final Project (Module 4)**: Student-designed research or application project using advanced neural/transformer-based methods.

1 Mini-Project (Module 3)

Objective

The Mini-Project consolidates fundamental NLP concepts covered in the first module of the course. It should combine preprocessing, representation, and modeling steps to form an end-to-end pipeline.

Structure

- Work individually or in pairs.
- Select one of the guided tracks below or propose a similar variant.
- Duration: 2-4 weeks (Weeks 6–8).
- Deliverables: GitHub repository, short paper (2–3 pages), 5-minute presentation.

Suggested Tracks (Idea)

1. **Classical NLP Pipeline:** Tokenization → TF–IDF → Logistic Regression or Feedforward Neural Network.
   Goal: Compare preprocessing choices and analyze their impact on performance.
   Dataset: IMDb reviews, AG News, or a Russian sentiment corpus.

2. **Embedding-Based Classifier:** Use word2vec or GloVe embeddings and train Feedforward Neural Network classifier.
   Goal: Compare pretrained vs. locally trained embeddings on small data.
   Dataset: News or Wikipedia paragraphs.

3. **Sequence Modeling (optional advanced):** Train an more complex (not being taught during the lecture) on character- or word-level language modeling.
   Goal: Explore overfitting and perplexity as a diagnostic tool.

4. **Quantitative Study:** Analyze tokenizer or language model errors across languages or domains.
   Goal: Demonstrate ability to collect and interpret linguistic data.

Grading (100 points)

Component	Points
Technical correctness & implementation	30
Experimental design & analysis	25
Insight and interpretation	20
Code organization & reproducibility	15
Presentation & clarity of report	10

Timeline

Week	Milestone	Deliverable
6	Topic selection	1-paragraph plan approved by instructor
7	Preliminary results	short check-in during lab
8	Submission & presentation	repository + short paper

2 Final Project (Module 4)

Objective

The Final Project is a capstone research or applied project, designed and executed by students. It should demonstrate mastery of modern NLP methods, including transformers, retrieval, or alignment approaches (if time permits).

Structure

- Work in teams of 1–3 students (2 recommended).

- Duration: Weeks 3–6 (6 weeks).
- Proposal approval required (Week 8 deadline). Sent an email to sshalileh@hse.ru, indicate title, dataset(s) and objective in one paragraph.
- Deliverables: GitHub repository, 4–6 page paper, 10-minute demo presentation.

Example Project Ideas

1. **Domain-Specific Fine-Tuning:** Adapt BERT or RoBERTa to a domain corpus (finance, legal, health). Evaluate with domain-specific metrics.
2. **Cross-Lingual Transfer:** Compare multilingual transformers (mBERT, XLM-R) for NER or sentiment across languages.
3. **Retrieval-Augmented QA:** Implement a small RAG pipeline using dense or sparse retrievers; evaluate latency vs. accuracy.
4. **Instruction-Tuned Models:** Perform small-scale SFT or DPO on a compact model; design custom evaluation rubric for helpfulness.
5. **Interpretability Study:** Analyze token-level saliency or probing on pretrained LMs; visualize hidden representations.
6. **Bias and Fairness Analysis:** Examine demographic or topical bias in model outputs using template-based generation.
7. **Low-Resource Adaptation:** Explore parameter-efficient fine-tuning (LoRA, prefix-tuning) for small corpora.
8. **NLP Systems Project:** Build an applied pipeline (chatbot, text summarizer, domain QA) integrating modern APIs or models.

Grading Rubric (100 points)

Component	Points
Problem definition & clarity	10
Data & experimental setup	20
Technical correctness & methodology	25
Analysis & interpretation	25
Writing & presentation quality	15
Originality & insight	5

General Notes

- Reports should follow the ACL two-column format (LaTeX or Overleaf template).
- Code must be reproducible and accompanied by a clear `README`.

- Datasets must be publicly available or properly licensed.
- Collaborations across teams are permitted for discussion but not code sharing.
- Outstanding projects may be invited for research paper submissions or lab collaboration.